On Job Training-2

1.Select any one job role of his/her interest in an organization for next 4 weeks and submit a report to the training supervisor and copy to cohort owner focusing on:

Job Role: AI/ML Intern (Backend Development & Automation)

This project aims to develop a **machine learning-based web application for Agriculture Drought Prediction**, which uses environmental parameters such as rainfall, soil moisture, temperature, humidity, and wind speed to predict whether a region is facing drought conditions or not.

The system integrates **frontend design, backend processing, and prediction logic** to create a complete application. The work carried out during the internship involved designing login and dashboard interfaces, implementing backend logic using Flask, and generating visual outputs such as GIFs and donut charts.

I. Application of Skills and Technical Knowledge

During the internship, I applied various technical skills and tools to develop the drought prediction system and perform OJT tasks.

- **Programming Language: Python**

Python was used for backend logic, prediction handling, and integrating different components of the system.

- **Libraries Used: Pandas,Scikit-learn**

Pandas
Used for data handling and preprocessing such as organizing input data, cleaning, and formatting.
Scikit-learn (sklearn)
Used for implementing the machine learning model for drought prediction

- **Backend Framework: Flask**

Flask was used to:

- Handle user requests

- Process input data
- Generate prediction output
- Connect frontend with backend

➢ Frontend Technologies: HTML, CSS

- Designed **login page** and **dashboard interface**
- Created input forms for environmental parameters
- Used CSS for styling and improving user experience

➢ Data Visualization

- Implemented **donut chart** for probability display
- Used visual outputs:
- Rainfall GIF (No Drought)
- Dry land GIF (Drought)

➢ API Development (OJT Learning)

- Learned invoice system concepts
- Developed APIs:

  - Invoice API
  - Payment API
  - Credit Memo API
  - Aging Report API
  - Ledger API

➢ Excel Data Processing

- Extracted invoice data into Excel
- Performed:

  - Data merging
  - Record matching
  - Data cleaning

The integration of these technologies helped in building a complete system combining **AI/ML, frontend, backend, and data processing**.

II. Performance on Assigned Tasks and Project

The assigned tasks were completed in a structured and systematic manner throughout the internship.

- Developed a **complete drought prediction system** with frontend and backend integration
- Successfully handled user inputs and generated accurate outputs
- Implemented visual representation using charts and GIFs
- Designed clean and user-friendly UI

In addition:

- Developed and tested multiple invoice-related APIs
- Performed Excel data processing tasks such as merging and matching records
- Ensured accuracy and consistency in

Value Addition to the Organization

The internship contributed value to the organization in multiple ways:

- The **drought prediction system** provides an intelligent solution for analyzing environmental conditions
- Helps in **decision-making support** for agriculture-related insights
- Reduces manual effort by automating prediction and output generation

From a technical perspective:

- API development improved backend system capabilities
- Excel data processing improved data organization and accuracy
- Integration of multiple technologies improved workflow efficiency

III. Job Role Understanding and Relevance

The role undertaken during the internship aligns with that of an **AI/ML and Backend Development Intern**.

Responsibilities included:

- Frontend interface design
- Backend development using Flask
- Machine learning logic implementation
- API development
- Data processing and validation
- Debugging and optimization

This role helped in understanding how theoretical knowledge can be applied in real-world applications.

Use Case -2

2. Document a Use Case on a Task where he/she is working as an intern

Use Case Documentation

2.1 Use Case Title

Agriculture Drought Prediction System

2.2 Description

The system predicts whether a region is experiencing drought conditions based on environmental input parameters. It provides both prediction results and visual outputs for better understanding.

2.3 Actors

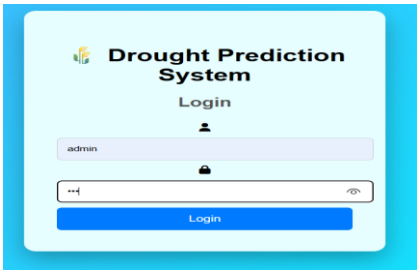
- System
- User (Admin/Organization)

2.4 Inputs

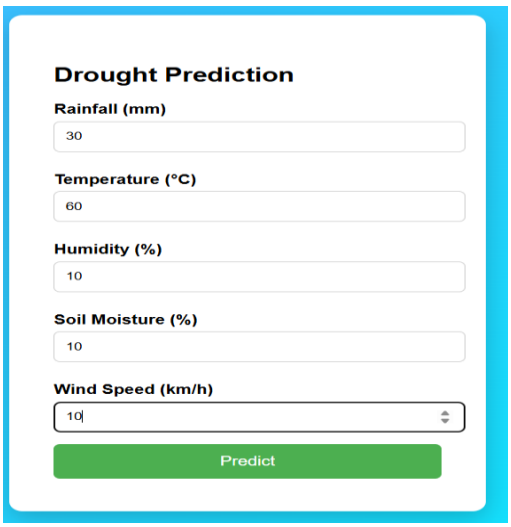
- Rainfall
- Soil Moisture
- Temperature
- Humidity
- Wind Speed

2.5 Process

- User logs into the system using login page



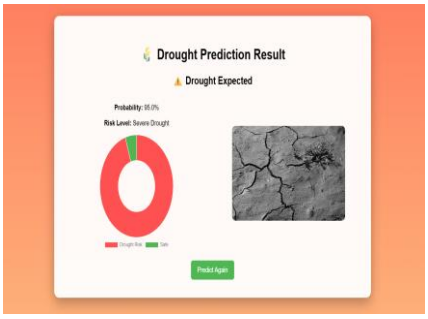
- Dashboard is displayed

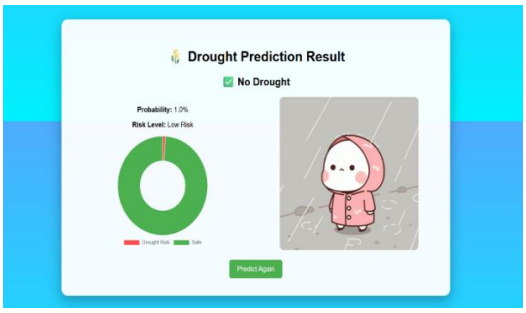


- User enters environmental parameters
- Inputs are sent to backend (Flask)
- Prediction logic processes the data
- System predicts:
  - Drought
  - No Drought
- Result is displayed with:
  - GIF (rainfall or dry land)
  - Donut chart (probability)

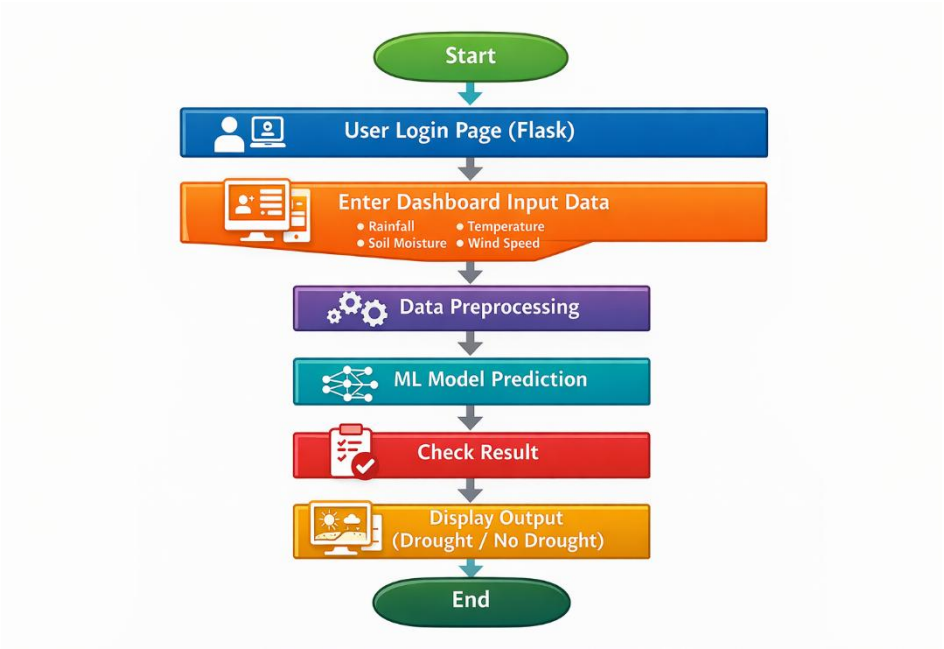
2.6 Output

- Prediction result (Drought / No Drought)
- Probability visualization (Donut Chart)
- Visual output (GIF)





2.7 Diagram



The above diagram represents the workflow of the Agriculture Drought Prediction System. The process begins with the user logging into the system through the Flask-based login page.The system then checks the result and displays the final output as either **Drought** or **No Drought** along with visual representation. The process ends after displaying the result to the user.

2.8 Advantages

- Provides real-time prediction
- User-friendly interface
- Visual representation improves understanding
- Reduces manual analysis effort

2.9 Limitations

- Accuracy depends on input values
- Basic model implementation
- Limited dataset scope

2.10 Future Enhancements

- Improve model accuracy using advanced ML algorithms
- Integrate real-time weather data APIs

- Add more environmental parameters
- Enhance UI and visualization

Conclusion

In conclusion, the Agriculture Drought Prediction System demonstrates the effective application of AI/ML techniques in solving real-world problems. The system successfully integrates frontend, backend, and prediction logic to provide meaningful outputs.

In addition, the internship provided strong exposure to API development and data processing tasks, enhancing technical and problem-solving skills.

With further improvements, this system has the potential to support better decision-making in agriculture and environmental analysis.